3D WEBSOFT INC. FOUNDATION

FUTURE TECH SCHOOL PROGRAM

Faculty Name :

Course Duration : 5 Months

Classes : Offline & Online

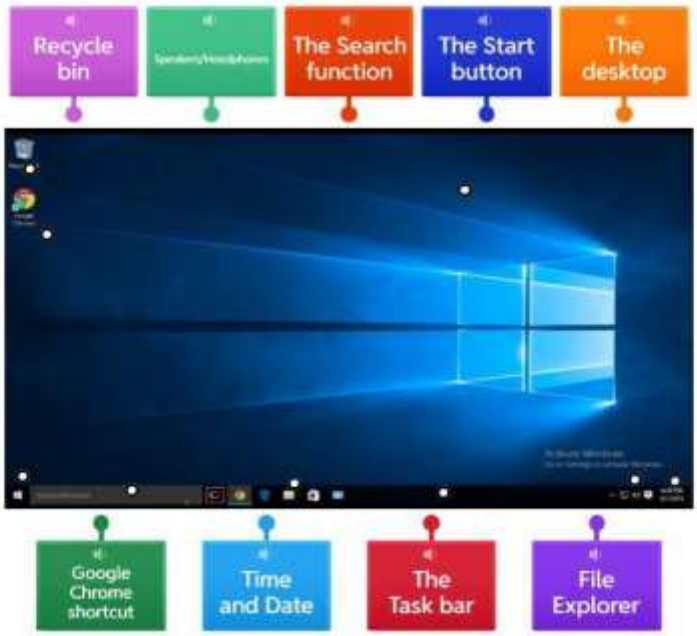
Daily Teaching : 2 Hours

Total Classes : 100

Level 2 – Young Technologist (7th–8th Standard)

Class 5: Operating System Basics







Learning Objectives

By the end of this lesson, students will be able to:

- Understand what an Operating System (OS) is.
- Explain why an Operating System is important.
- Identify the main functions of an Operating System.

- Recognize different types of Operating Systems.
- Identify common Operating Systems used on computers and mobile devices.
- Perform basic tasks using the Windows Operating System.

Introduction

Imagine a computer without an Operating System.

- The monitor cannot display anything.
- The keyboard and mouse will not work properly.
- You cannot open games or applications.

An **Operating System (OS)** is the **most important software** that allows the computer to work.

It acts as a **bridge between the user and the computer hardware**.

What is an Operating System?

An **Operating System (OS)** is **system software** that manages the computer's hardware and software resources and allows users to interact with the computer.

Simple Definition

An Operating System is the **manager of the computer**. It controls all the computer's activities and helps different programs work together.

Why is an Operating System Important?

Without an Operating System:

- The computer cannot start.
- Programs cannot run.
- Files cannot be managed.
- Hardware devices cannot communicate with the computer.

Everyday Example

When you turn on a computer:

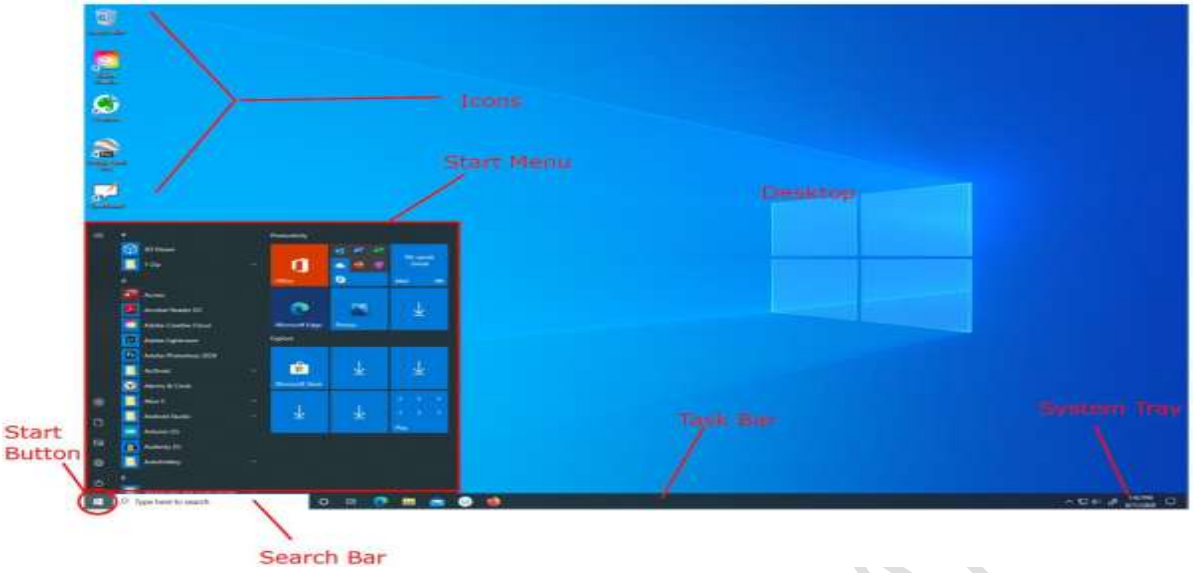
1. The Operating System starts.

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2. The desktop appears.
3. You can open applications such as Paint, Calculator, or a web browser.

Functions of an Operating System




Icons
Start Menu
Desktop
Start Button
Task Bar
System Tray
Search Bar


Core Responsibilities of an Operating System (OS)
Memory Management
Process Management
File Management
Device Management
Device (I/O) Management
Performance Management
Security & Protection
User Interface
User Interface

1. Booting the Computer

The Operating System starts the computer when you press the power button.

Example: Windows loads and displays the desktop.

2. Managing Hardware

The OS controls hardware devices such as:

- Keyboard
- Mouse
- Monitor
- Printer
- Speakers

3. Running Applications

The OS allows us to open and use applications.

Examples:

- Paint
- Calculator
- Microsoft Word
- Python
- Google Chrome

4. File Management

The OS helps us:

- Create folders
- Save files
- Copy files
- Move files
- Delete files

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5. Memory Management

The OS manages the computer's memory so that different programs can run smoothly.

6. Device Management

The OS communicates with printers, webcams, USB drives, and other connected devices.

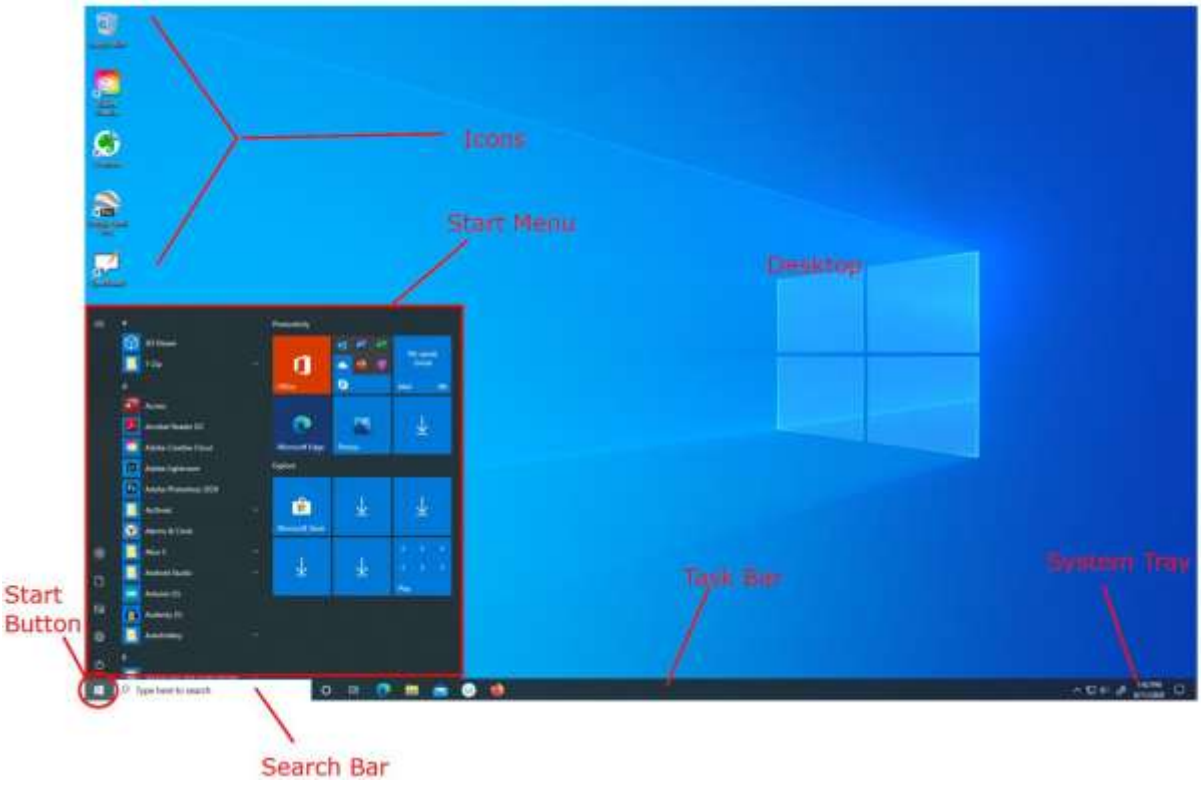
7. Security

The OS helps protect the computer using:

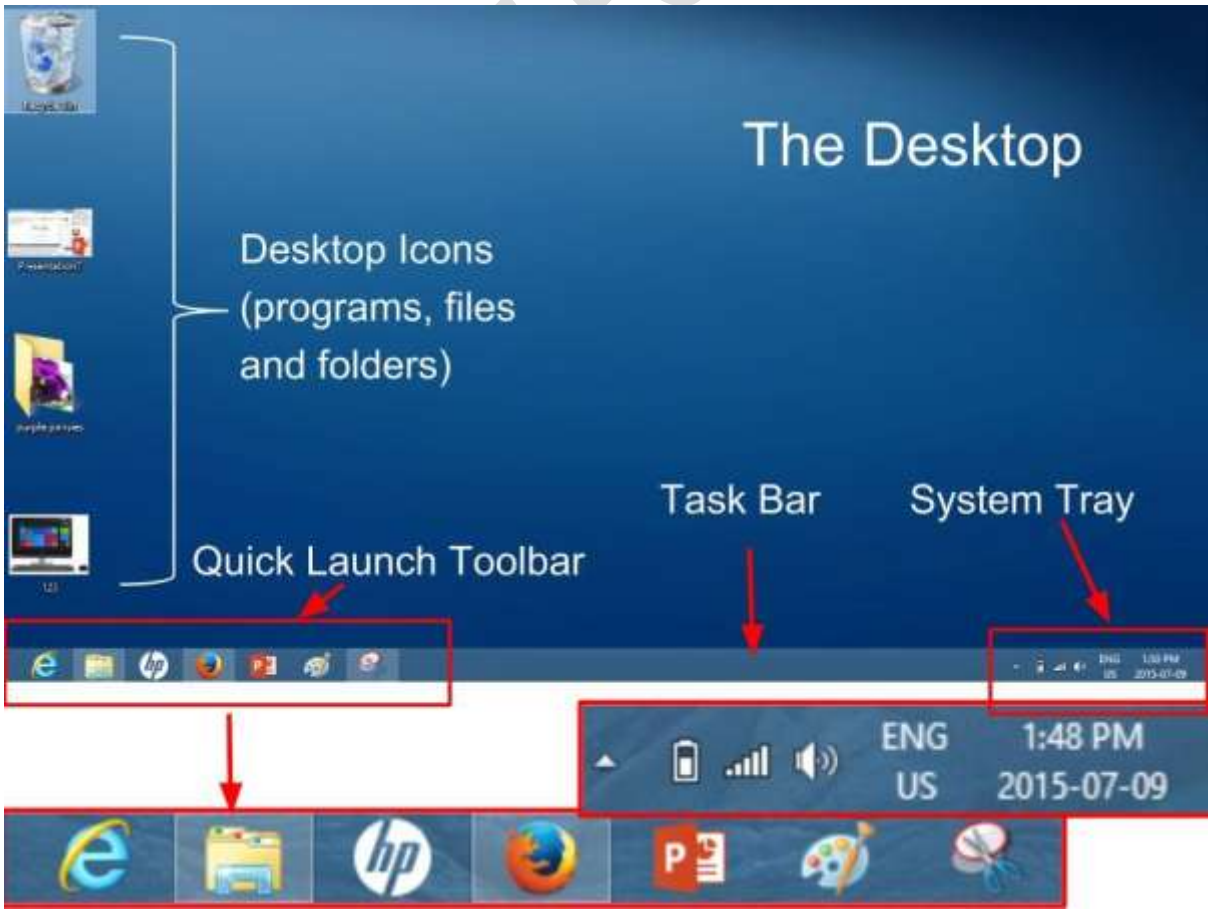
- User accounts
- Passwords
- Security updates
- Built-in protection tools

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Parts of the Windows Desktop




Tiles
user Account
Start button
Notification Area
Start Menu
Pinned App
desktop
Search box
Task Bar
System Icons
Frequent used App


The Desktop
Desktop Icons (programs, files and folders)
Task Bar
System Tray
Quick Launch Toolbar
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1. Desktop

The main screen displayed after the computer starts.

2. Icons

Small pictures that represent files, folders, or applications.

Examples:

- Recycle Bin
- This PC
- Paint

3. Taskbar

Usually located at the bottom of the screen.

It shows:

- Running applications
- Start Menu
- Clock
- Notifications

4. Start Menu

Used to:

- Open applications
- Search files
- Shut down the computer
- Access settings

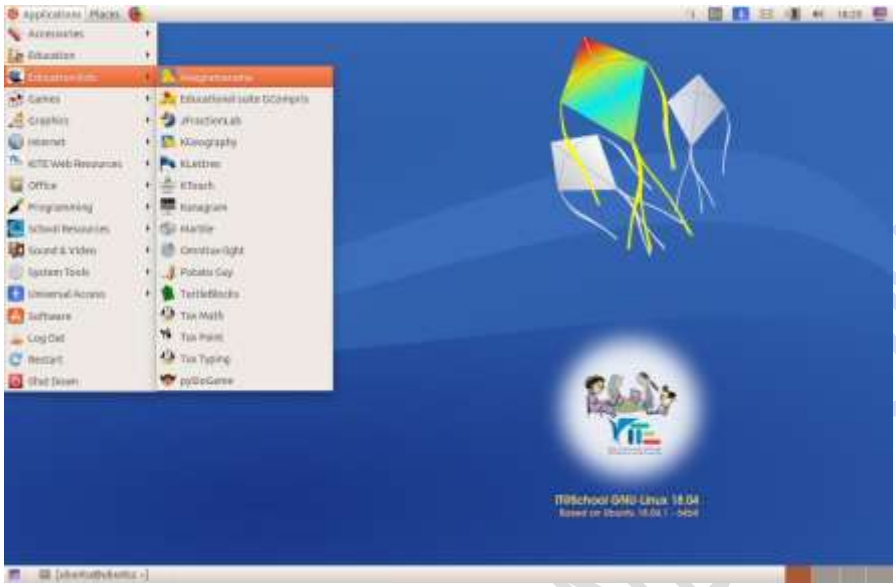
5. Recycle Bin

Stores deleted files temporarily.

Files can be restored if needed before permanently deleting them.

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Types of Operating Systems



1. Windows

- Developed by Microsoft.
- Popular in schools, homes, and offices.
- User-friendly interface.

2. macOS

- Developed by Apple.
- Used on Mac computers.

3. Linux

- Free and open-source.
- Popular with programmers and servers.

4. Android

- Used on most smartphones and tablets.

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5. iOS

- Developed by Apple.
- Used on iPhones and iPads.

Operating System vs Application Software

Operating System	Application Software
Starts the computer	Performs specific tasks
Controls hardware	Helps users create documents, browse the internet, draw, etc.
Example: Windows, Linux	Example: Paint, Word, Chrome, Python

Real-Life Examples

Device	Operating System
Desktop Computer	Windows, Linux
Laptop	Windows, macOS
Android Phone	Android
iPhone	iOS
Tablet	Android, iPadOS

Advantages of an Operating System

- Makes computers easy to use.
- Runs applications smoothly.
- Manages files and folders.
- Supports multiple hardware devices.
- Provides security features.
- Improves overall computer performance.

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Computer Lab Rules

- Start the computer only when instructed.
- Do not change system settings without permission.
- Save your work before shutting down.
- Shut down the computer properly.
- Handle the keyboard and mouse carefully.

Practical Activities

Practical Activity 1 – Explore the Windows Desktop

Objective: Identify the main parts of the desktop.

Instructions

1. Turn on the computer.
2. Observe the desktop.
3. Identify:
    - Desktop
    - Icons
    - Taskbar
    - Start Menu
    - Recycle Bin
4. Click the **Start Menu** and search for **Calculator**.
5. Open Calculator and then close it.

Practical Activity 2 – Open an Application

Instructions

1. Click the **Start Menu**.
2. Open **Paint**.

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3. Draw a smiley face.
4. Save the file as **Smiley.png**.
5. Close Paint.

Practical Activity 3 – Create a Folder

Instructions

1. Right-click on the desktop.
2. Select **New → Folder**.
3. Name it **My Projects**.
4. Open the folder.
5. Close it.

Classroom Activity

Windows Part Function

Desktop	Main screen
Start Menu	Open applications
Taskbar	Shows running programs
Recycle Bin	Stores deleted files
Icons	Represent apps and files

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Quick Quiz

1. What does OS stand for?

A. Open Software

B. Operating System ✅

C. Online System

D. Office System

2. Which software starts the computer?

A. Paint

B. Calculator

C. Operating System ✅

D. Python

3. Which Operating System is commonly used on desktop computers?

A. Windows ✅

B. Paint

C. Chrome

D. PowerPoint

4. Which part of the Windows desktop stores deleted files?

A. Start Menu

B. Taskbar

C. Recycle Bin ✅

D. Folder

5. Which Operating System is commonly used on Android phones?

A. Windows

B. Android ✅

C. Linux

D. macOS

Home Assignment

1. Write the definition of an Operating System.
2. List **five functions** of an Operating System.

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3. Draw and label the Windows desktop (Desktop, Icons, Taskbar, Start Menu, Recycle Bin).
4. Name **five Operating Systems** used on computers or mobile devices.
5. Ask a family member which Operating System they use on their phone or computer.

Lesson Summary

In this lesson, students learned:

- What an Operating System is.
- Why an Operating System is important.
- The main functions of an Operating System.
- The parts of the Windows desktop.
- Different Operating Systems such as Windows, macOS, Linux, Android, and iOS.
- Basic Windows practical skills like opening applications and creating folders.

Learning Outcomes

By the end of this lesson, students will be able to:

- Define an Operating System.
- Explain why an Operating System is essential.
- Identify the main functions of an Operating System.
- Recognize the parts of the Windows desktop.
- Name common Operating Systems used on computers and mobile devices.
- Perform basic Windows tasks such as opening applications, creating folders, and shutting down the computer correctly.

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